

# Car Price Prediction using RANDOM FOREST

```
import pandas as pd
```

## *Load Dataset from Local directory*

```
from google.colab import files  
uploaded = files.upload()
```

## *Load Dataset*

```
dataset = pd.read_csv('dataset.csv')  
dataset = dataset.drop(['car_ID'],axis=1)
```

## *Summarize Dataset*

```
print(dataset.shape)  
print(dataset.head(5))
```

## *Splitting Dataset into X & Y*

### *This X contains Both Numerical & Text Data*

```
Xdata = dataset.drop('price',axis='columns')  
numericalCols=Xdata.select_dtypes(exclude=['object']).columns  
X=Xdata[numericalCols]  
X  
  
Y = dataset['price']  
Y
```

## *Scaling the Independent Variables (Features)*

```
from sklearn.preprocessing import scale  
cols = X.columns  
X = pd.DataFrame(scale(X))  
X.columns = cols  
X
```

## *Splitting Dataset into Train & Test*

```
from sklearn.model_selection import train_test_split  
x_train,x_test,y_train,y_test =  
train_test_split(X,Y,test_size=0.20,random_state=0)
```

### *Training using Random Forest*

```
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor()
model.fit(x_train, y_train)
```

### *Evaluating Model*

```
ypred = model.predict(x_test)

from sklearn.metrics import r2_score
r2score = r2_score(y_test,ypred)
print("R2Score",r2score*100)
```